

5.3.1 The Fundamental Theorem Worksheet

1. Compute, if possible, the following definite integrals.

a) $\int_2^4 (4x - 3) dx$

b) $\int_0^2 (6t^2 + 5e^t + 2) dt$

c) $\int_{-2}^3 \frac{8(x-2) + x}{(x-2)} dx$

d) $\int_1^4 \frac{18\sqrt{x} - x^3}{3x} dx$

e) $\int_1^9 \frac{1}{2x} dx$

f) The area under the parabola $y = x^2$ from 0 to 1.

2. What is wrong with the following calculation?

$$\int_0^2 \frac{-1}{(x-1)^2} dx = \frac{1}{(x-1)} \Big|_0^2 = \frac{1}{1} - \left(\frac{1}{-1} \right) = 1 + 1 = 2$$

3. Suppose $w(x) = \int_5^{x^2} \frac{\ln(t+6)}{t} dt$. Find $w'(x)$.

4. Suppose $H(x) = \int_{\sqrt{x}}^{10} 6 \left(t^4 - \frac{1}{t} \right)^{3/7} dt$. Find $H'(x)$.

5. Suppose $p(x) = \int_8^{x^3-75x} \frac{1}{t^4+100} dt$. Determine each interval upon which the graph of $p(x)$ is decreasing.

6. Find the average value of the function $f(x) = 1 + x^2$ on the interval $[-1, 2]$.

7. Suppose $F(x)$ is a polynomial with $F'(x) = f(x)$. Given that:

$F(0) = 2$, $F(2) = 8$, $F(4) = 28$, $F(6) = 68$, and $F(8) = 42$ find the average value of $f(x)$ on the interval $[2, 6]$.

8. Suppose $w(x) = \int_{10}^{x^2} (t-4)(t+1)^6 dt$. Determine all the intervals upon which $w(x)$ is increasing.